

06.11.25

8 пр.

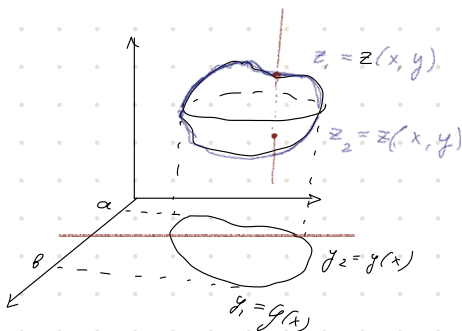
$$\begin{aligned}
 a) \int_{-1}^3 dx \int_0^x dy \int_0^{\sqrt{x^2-y^2}} (\sqrt{x^2-y^2} + z) dz &= \int_{-1}^3 dx \int_0^x dy \sqrt{x^2-y^2} + \frac{z^2}{2} \Big|_0^{\sqrt{x^2-y^2}} = \\
 &= \int_{-1}^3 dx \int_0^x \left(\sqrt{x^2-y^2} + \frac{x^2-y^2}{2} \right) dy = \int_{-1}^3 dx \left[\frac{y}{2} \sqrt{x^2-y^2} - \frac{y^3}{3} + \frac{x^2 y}{2} - \frac{y^3}{3} \right]_0^x = \\
 &= \int_{-1}^3 dx \left[\frac{9}{8} x^3 - \frac{1}{6} x^3 \right] dx = \int_{-1}^3 \frac{5}{6} x^3 dx = \frac{5}{6} \cdot \frac{x^4}{4} \Big|_{-1}^3 = \frac{5 \cdot 81}{24} - \frac{5 \cdot 1}{24} = \dots
 \end{aligned}$$

$$\begin{aligned}
 б) \int_0^1 dx \int_0^{2x} dy \int_0^{\sqrt{xy}} x^3 y^2 z^2 dz &= \int_0^1 dx \int_0^{2x} dy \frac{x^3 y^2 z^3}{3} \Big|_0^{\sqrt{xy}} = \\
 &= \int_0^1 dx \int_0^{2x} dy \frac{x^4 y^3}{2} = \int_0^1 dx \frac{x^4 y^4}{8} \Big|_0^{2x} = \int_0^1 \frac{x^4 \cdot 16 x^4}{8} dx = \frac{2 x^9}{9} \Big|_0^1 = \frac{2}{9}.
 \end{aligned}$$

= 20

Тройной интеграл

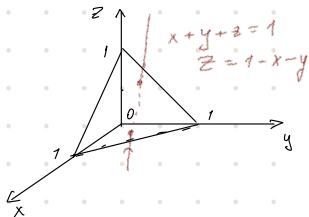
интегр. по T-какому-то телу



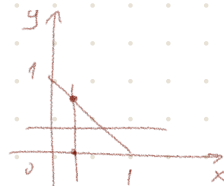
$$\begin{aligned}
 \iiint_T f(x, y, z) dx dy dz &= \\
 &= \int_a^b dx \int_{y_1(x)}^{y_2(x)} dy \int_{z_1(x, y)}^{z_2(x, y)} f dz
 \end{aligned}$$

② Вычисл. тройной интеграл $\iiint_T z dx dy dz$

если T: $z=0, x=0, y=0, x+y+z=1$



$$\begin{aligned}
 x+y+z &= 1 \\
 x=0: \quad z &= 1-y \\
 y=0: \quad z &= 1-x
 \end{aligned}$$



$$\int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} z dz = \int_0^1 dx \int_0^{1-x} \frac{(1-x-y)^2}{2} dy =$$

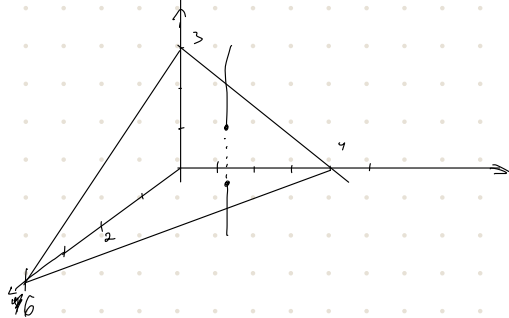
$$= \int_0^1 dx \left(-\frac{(1-x-y)^3}{3} \Big|_0^{1-x} \right) = \int_0^1 dx \left(-\frac{(1-x)^3}{3} \right) \Big|_0^1 =$$

$$= -\int_0^1 \frac{(1-x)^3}{3} d(1-x) = -\frac{(1-x)^4}{24} \Big|_0^1 = \frac{1}{24}$$

③ Расставь пределы интегрир. в

$$\iiint_T f(x, y, z) dx dy dz,$$

если T -тетраэдр $2x + 3y + 4z = 12$, $x=0$, $y=0$, $z=0$



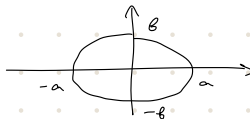
$$x=0: z = \frac{12-3y}{4} = 3 - \frac{3}{4}y$$

$$y=0: z = \frac{12-2x}{4} = 3 - \frac{1}{2}x$$

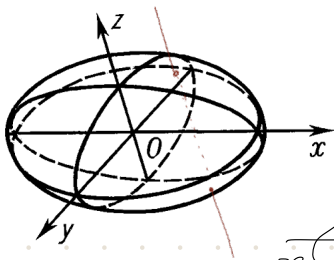
$$\int_0^6 dx \int_0^{4-\frac{2}{3}x} dy \int_0^{12-3y-2x} f(x, y, z) dz$$

④ $\rightarrow$ T -внутр. эллипсоида

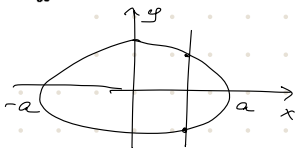
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$



$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

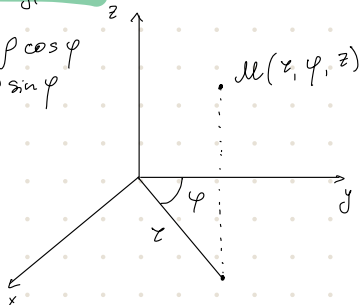


$$\int_{-a}^a dx \int_{-\sqrt{b^2 - \frac{x^2}{a^2}}}^{\sqrt{b^2 - \frac{x^2}{a^2}}} dy \int_{-\sqrt{c^2 - \frac{c^2 x^2}{a^2} - \frac{c^2 y^2}{b^2}}}^{\sqrt{c^2 - \frac{c^2 x^2}{a^2} - \frac{c^2 y^2}{b^2}}} f(x, y, z) dz$$

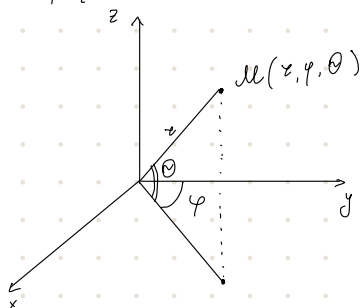


цилиндрич. с.к.

$$\begin{aligned}x &= \rho \cos \varphi \\y &= \rho \sin \varphi \\z &= z\end{aligned}$$



сферич. с.к.



5) Перейти к цилиндрическим коорд. и вычисл.

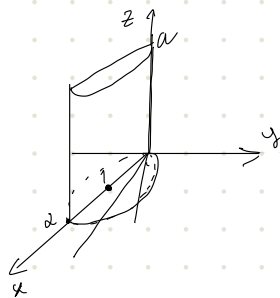
$$\iiint_T z \sqrt{x^2 + y^2} dx dy dz$$

$$T: \begin{aligned}0 &\leq x \leq 2, \\0 &\leq y \leq \sqrt{2x - x^2} \\0 &\leq z \leq a\end{aligned}$$

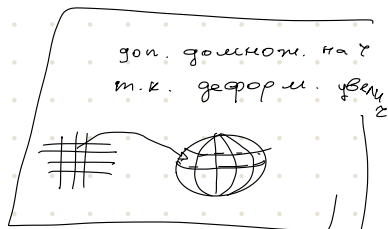
кажд. вычисл. одинаков, поэтому равен ρ

$$\begin{aligned}y &= \sqrt{2x - x^2} \\y^2 &= 2x - x^2 \\x^2 + y^2 - 2x &= 0 \\(x-1)^2 + y^2 &= 1\end{aligned}$$

$$|J| = \rho$$



$$\begin{cases}x = \rho \cos \varphi \\y = \rho \sin \varphi \\z = z\end{cases}$$



$$z \sqrt{x^2 + y^2} = z \sqrt{z^2 (\cos^2 \varphi + \sin^2 \varphi)} = z^2$$

$$\int_0^a z \int_0^{2\pi} d\varphi \int_0^{2\cos \varphi} z \cdot z dz$$

Получен с конца:

$$\begin{aligned}z^2 \cos^2 \varphi + z^2 \sin^2 \varphi - 2z^2 \cos \varphi &= 0 \\z^2 &= 2z \cos \varphi \\z &= 2 \cos \varphi\end{aligned}$$

$$\begin{aligned}&\int_0^a dz \int_0^{2\pi} d\varphi \int_0^{2\cos \varphi} z^2 dz = \int_0^a dz \int_0^{2\pi} d\varphi \int_0^{2\cos \varphi} z^3 dz = \\&= \int_0^a dz \int_0^{2\pi} \left(\frac{z^4}{4} \right)_{z=0}^{z=2\cos \varphi} d\varphi = \int_0^a dz \int_0^{2\pi} \frac{16 \cos^4 \varphi}{4} d\varphi =\end{aligned}$$